



5.2 - Marginal and Conditional Odds Ratios

Marginal Odds Ratios

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Marginal odds ratios are odds ratios between two variables in the marginal table and can be used to test for marginal independence between two variables while ignoring the third.

For example, for the XY margin, where $\mu_{ij\cdot}$ denotes the expected count of individuals with $X = i$ and $Y = j$ in the marginal table obtained by summing over Z , the **marginal odds ratio** is

$$\theta_{XY} = \frac{\mu_{11\cdot}\mu_{22\cdot}}{\mu_{12\cdot}\mu_{21\cdot}}$$

And the estimate of this from the admission data would be

$$\hat{\theta}_{XY} = \frac{1198 \cdot 1278}{1493 \cdot 557} = 1.84$$

Thus, if we aggregate values over all departments, the odds that a male is admitted are an estimated 1.84 times as high as the odds that a female is admitted. And if we were to calculate a suitable confidence interval (e.g., 95%), it would not include 1, indicating that the odds for males is significantly higher.

Conditional Odds Ratios

Conditional Odds Ratios

Conditional odds ratios are odds ratios between two variables for fixed levels of the third variable and allow us to test for conditional independence of two variables, given the third.

For example, for the fixed level $Z = k$, the conditional odds ratio between X and Y is

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}}$$

There are as many such conditional odds ratios as there are levels of the conditional variable, and each can be estimated from the corresponding conditional or partial table between X and Y , given $Z = k$. For the first two departments of the admission data, the estimated conditional odds ratios between sex and admission status would be

$$\hat{\theta}_{XY(Z=1)} = \frac{512 \cdot 19}{89 \cdot 313} = 0.35$$

and

$$\hat{\theta}_{XY(Z=2)} = \frac{353 \cdot 8}{17 \cdot 207} = 0.80$$

That is, if we restrict our attention to Department A only, the odds that a male is admitted are an estimated 0.35 times as high as the odds that a female is admitted. Or, equivalently, the odds that a female is admitted are an estimated $1/0.35 = 2.86$ times that for males. Likewise, the odds of being admitted are higher for females if we restrict our attention to Department B.

We are calculating the odds ratios for the various partial tables of the larger table and can use them to test the conditional independence of X and Y , given Z . If $\theta_{XY(k)} \neq 1$ for at least one level of Z (at least one k), it follows that X and Y are conditionally associated. We will learn more about this, but for now, let's utilize our knowledge of two-way tables to do some preliminary analysis.

NOTE: Marginal association can be very different from conditional association. That is, marginal and conditional odds ratios do NOT need to be equal. In fact, sometimes they may lead to quite the opposite conclusions!

Using what we know about 2×2 tables and tests for association, we can compare the marginal and conditional odds ratios for our example and measure evidence for their significance. What do they tell us about the relationships among these variables?

1. Let's look at using the SAS program file [berkeley.sas](#) (Full output: [berkeley.SAS Output](#)).

```
/* Analysis of a 3-way table Berkeley admissions data using PROC FREQ */
options nocenter nodate nonumber linesize=72;
data berkeley;
  input D $ S $ A $ count;
cards;
DeptA Male Reject 313
DeptA Male Accept 512
DeptA Female Reject 19
DeptA Female Accept 89
DeptB Male Reject 287
DeptB Male Accept 353
DeptB Female Reject 8
DeptB Female Accept 17
DeptC Male Reject 285
DeptC Male Accept 120
DeptC Female Reject 191
DeptC Female Accept 282
DeptD Male Reject 279
DeptD Male Accept 138
DeptD Female Reject 244
DeptD Female Accept 131
DeptE Male Reject 138
DeptE Male Accept 53
DeptE Female Reject 299
DeptE Female Accept 94
DeptF Male Reject 351
DeptF Male Accept 12
DeptF Female Reject 317
DeptF Female Accept 24
;
/*analysis of the three-way table including CMH test*/
proc freq data=berkeley order=data;
weight count;
tables D*S*A /cm chisq relrisk expected nocol norow;
tables S*A/chisq relrisk;
run;
```

The tables command is where we can specify which variables to tabulate; those that are omitted are summed over (marginalized). For example,

tables S*A/chisq all nocol nopct;

will create a marginal table of sex and admission status, and compute all the relevant statistics for this 2×2 table (see below). To get the partial tables and analyses of sex and admissions status for each department, we can run the following line:

tables D*S*A /chisq cmh nocol nopct;

We will discuss the CMH option later. In PROC FREQ, the partial tables will be created given the levels of the first variable you specify when creating a three-way table. We can see the full output of this program in [berkeley.SAS Output](#).

Statistical Inference

Marginal Independence

Let's first look at the marginal table of sex and admission status, while ignoring departments. As we calculated earlier, the point estimate of the odds-ratio is 1.84. That is, the odds of admission for males are an estimated 1.84 times as high as that for females, and it can be shown to be statistically significant based on the 95% confidence interval or a chi-square test of independence. However, keep in mind that we ignored the department information here. A more precise statement would be to say that sex and admission status are marginally associated.

Frequency Percent Row Pct Col Pct	Table of S by A			
	S	A		
		Reject	Accept	Total
Male		1492	1199	2691
		32.97	26.49	49.46
		55.44	44.56	
		53.86	68.28	
Female		1278	557	1835
		28.24	12.31	40.54
		69.65	30.35	
		46.14	31.72	
Total		2770	1756	4526
		61.20	38.80	100.00

Statistics for Table of S by A			
Statistic	DF	Value	Prob
Chi-Square	1	92.6704	<.0001
Likelihood Ratio Chi-Square	1	93.8232	<.0001
Continuity Adj. Chi-Square	1	92.0733	<.0001
Mantel-Haenszel Chi-Square	1	92.6499	<.0001
Phi Coefficient		-0.1431	
Contingency Coefficient		0.1416	
Cramer's V		-0.1431	

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	0.5423	0.4785	0.6147
Relative Risk (Column 1)	0.7961	0.7008	0.8330
Relative Risk (Column 2)	1.4679	1.3535	1.5919

Conditional Independence

Now consider the point estimates of odds ratios when we control for the department, which uses conditional odds-ratios (see Sec. 5.2.) Given that the individuals are applying to Department A, the odds of male admission are 0.35 times as high as the odds of female admission, and this is also significant at the 0.05 level (its 95% CI (0.2087, 0.5843) does not include the value 1). Other conditional odds ratios (Department B, C, etc.) are not significant, but it's interesting to note, nevertheless, that these conditional relationships do not have to be in the same direction as the marginal one. When they disagree, we have an example of Simpson's Paradox.

Frequency Expected Percent	Table 1 of S by A			
	S	Controlling for D=DeptA		
		Reject	Accept	Total
Male		313	512	825
		293.57	531.43	
		33.55	54.88	88.42
		38.431	60.969	
Female		19	89	108
		2.04	9.54	11.58
		332	601	933
		35.58	64.42	100.00

Statistics for Table 1 of S by A			
Controlling for D=DeptA			
Statistic	DF	Value	Prob
Chi-Square	1	17.2480	<.0001
Likelihood Ratio Chi-Square	1	19.0540	<.0001
Continuity Adj. Chi-Square	1	16.3718	<.0001
Mantel-Haenszel Chi-Square	1	17.2295	<.0001
Phi Coefficient		0.1360	
Contingency Coefficient		0.1347	
Cramer's V		0.1360	

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	2.8638	1.7112	4.7921
Relative Risk (Column 1)	2.1566	1.4206	3.2737
Relative Risk (Column 2)	0.7531	0.6799	0.8341

2. R users should open the [berkeley.R](#) file and its corresponding output file [berkeley.out](#). R will calculate the partial tables by the levels of the last variable in the array.

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```
#### Berkeley Admissions Example: a 2x2x6 table
#### 1st X=sex, Y=admission status, Z=department
library(survival)
library(vcd)
#### data available as an array in R
admit <- UCAdmissions
dimnames(admit) <- list(
  Admit=c("Admitted", "Rejected"),
  Sex=c("Male", "Female"),
  Dept=c("A", "B", "C", "D", "E", "F"))
admit
#### create a flat contingency table
ftable(admit, row.vars=c("Dept", "Sex"), col.vars="Admit")
#### conditional odds ratios
XY.Z <- oddsratio(admit) # log scale
exp(XY.Z$coef)
exp(confint(XY.Z))
plot(XY.Z)
#### Cochran-Mantel-Haenszel test of conditional independence
mantelhaen.test(admit)
mantelhaen.test(admit, correct=FALSE)
#### marginal tables and odds ratios
XY <- margin.table(admit, c(2,1))
ZY <- margin.table(admit, c(1,1))
XZ <- margin.table(admit, c(2,3))
oddsratio(XY, log=FALSE)
exp(confint(oddsratio(XY)))
chisq.test(XY, correct=FALSE)
#source(file.choose()) # choose breslowday.test_1.R
breslowday.test(admit)
```

We can also use `ftable()` function, e.g.,

```
ftable(admit, row.vars=c("Dept", "Sex"), col.vars="Admit")
```

to create flat tables. In this case, R essentially combines the departments and sexes into 12-row categories, resulting in a **12 × 2** representation of the original **2 × 2 × 6** table. To create a marginal table, we can use `margin.table()` to

```
margin.table(admit, c(2,1))
```

This function creates a marginal table of the second (sex) and the first (admission status) variables from the original array, which in this case puts the sexes as the rows and admission status groups as the columns.

Statistical Inference

Marginal Independence

Let's first look at the marginal table of sex and admission status, while ignoring departments. As we calculated earlier, the point estimate of the odds-ratio is 1.84. That is, the odds of admission for males are an estimated 1.84 times as high as that for females, and it can be shown to be statistically significant based on the 95% confidence interval or a chi-square test of independence. However, keep in mind that we ignored the department information here. A more precise statement would be to say that sex and admission status are marginally associated.

```
XY <- margin.table(admit, c(2,1))
chisq.test(XY, correct=FALSE)
```

Pearson's Chi-squared test

```
>data: XY
X-squared = 92.285, df = 1, p-value < 2.2e-16
```

Conditional Independence

Now consider the point estimates of odds ratios when we control for the department, which uses conditional odds-ratios (see Sec. 5.2.) Given that the individuals are applying to Department A, the odds of male admission are 0.35 times as high as the odds of female admission, and this is also significant at the 0.05 level (its 95% CI (0.2087, 0.5843) does not include the value 1). Other conditional odds ratios (Department B, C, etc.) are not significant, but it's interesting to note, nevertheless, that these conditional relationships do not have to be in the same direction as the marginal one. When they disagree, we have an example of **Simpson's Paradox**.

```
XY.Z <- oddsratio(admit) # log scale
exp(XY.Z$coef)
```

A	B	C	D	E	F
0.3492328	0.8825087	1.1338956	0.9212838	1.2216312	0.8278727
2.5 5	97.5 5				
A	0.2086756	0.5843954			
B	0.3403815	1.8928166			
C	0.8545328	1.5823686			
D	0.6863345	1.2366628			
E	0.8258748	1.8887848			
F	0.4552859	1.5856335			

Simpson's paradox

Simpson's paradox is the phenomenon that a pair of variables can have marginal association and partial (conditional) associations in opposite direction. Another way to think about this is that the nature and direction of association changes due to the presence or absence of a third (possibly confounding) variable.

In the simplest example, consider three binary variables, **X**, **Y**, **Z**. In the marginal table where we are ignoring the presence of **Z**, let

$P(Y = 1|X = 1) < P(Y = 1|X = 2)$

In the partial table, after we account for the presence of variable **Z**, let

$P(Y = 1|X = 1, Z = 1) > P(Y = 1|X = 2, Z = 1)$ and

$P(Y = 1|X = 1, Z = 2) > P(Y = 1|X = 2, Z = 2)$

In terms of odds ratios, marginal odds $\theta_{XY} < 1$, and partial odds $\theta_{XY(Z=1)} > 1$ and $\theta_{XY(Z=2)} > 1$.

These associations can also be captured in terms of models. Next, we explore more on different independence and association concepts that capture relationships between three categorical variables.

Additional Resources (optional)

Here is Dr. Jason Morton with a quick video explanation of what this paradox involves.

Simpson's Paradox



In addition, for those of you that would like to delve a little deeper into this, here is a link to "[Algebraic geometry of 2 × 2 contingency tables](#)" by Slavkovic and Fienberg. On page 17 of this document is a diagram of this paradox as well.

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